

Oluwatimilehin Ajibode

Linkedin: [linkedin.com/in/oluwatimilehin-timmy-ajibode](https://www.linkedin.com/in/oluwatimilehin-timmy-ajibode)

Github: github.com/timmyacy

Email : timmy_ajibode@hotmail.com

SUMMARY

Software engineer with four years of production experience building financial data pipelines and risk systems, and a recent MSc in Mathematical Finance with Distinction from the University of York. Focused on quantitative development at the intersection of low-latency engineering and derivatives pricing, with active projects in C++ covering FFT-based option pricing and Levy process models.

EDUCATION

- **University of York** York, United Kingdom
*MSc Mathematical Finance (Grade: **Distinction**)*
Sept 2024 – Sept 2025
 - Thesis: Multi-Level Monte Carlo methods for options pricing (**84%**)
- **Plymouth University** Plymouth, United Kingdom
BSc Computer Science with Professional Placement
Aug 2015 – Dec 2020

TECHNICAL SKILLS

- **Languages:** C++, Python, Java, SQL, TypeScript
- **Quantitative Libraries:** Pandas, NumPy, SciPy, Matplotlib, QuantLib
- **Platforms & Tools:** Kubernetes (CKAD), Docker, Jenkins, CI/CD, Git, PostgreSQL, MongoDB
- **Financial Modelling:** Stochastic Calculus (Itô's Lemma), Monte Carlo Methods, Numerical PDEs, Derivatives Pricing, Risk-Neutral Valuation, Carr-Madan FFT

EXPERIENCE

- **SAA Consultants** Plymouth, United Kingdom
Financial Software Engineer
Sept 2020 – Aug 2024
 - **Quantitative Analytics & Validation Engine:** Built a C++/Java framework validating millions of ISO 20022 and SWIFT financial messages daily across BACS, HMRC RTI/PAYE, and settlement pipelines. Identified a production bug where an optional field combination caused inconsistent notional scaling, producing incorrect values in downstream risk models.
 - **Low-Latency Data Processing Pipeline:** Engineered multi-threaded Java pipelines for real-time BACS payment and payroll transaction processing; Kubernetes containerisation of 7-8 REIMS microservices reduced deployment overhead and improved throughput.
 - **Analytics API Development:** Designed RESTful APIs (OpenAPI) giving risk analysts programmatic access to validated trade data and business rule logic, reducing manual reconciliation overhead across downstream systems.
 - **CI/CD Migration & Tooling:** Led migration of build and deployment pipelines from SVN to Jenkins with Git version control, containerising 7-8 REIMS microservices on Kubernetes and onboarding the team onto the new workflow.
- **Royal Cornwall Hospitals NHS Trust** Truro, United Kingdom
Honorary Research Assistant
Jun 2020 – Aug 2020
 - **Homecare System Redesign:** Technical requirements analysis and system architecture for a clinical homecare platform: NHS Spine/MPI integration, role-based access control, data validation for prescription and funding pipelines, and Blueteq API integration.

QUANTITATIVE MODELLING & SYSTEMS PROJECTS

- **C++ Algorithmic Trading System:** End-to-end trading system covering FIX session message parsing, L2 order book, UDP market data feed, position manager, volatility surface pricer, and live risk reporting. Low-latency design with clean separation between market data ingestion and risk layers.
- **C++ Levy Pricer:** FFT-based option pricing library implementing Carr-Madan with a from-scratch Cooley-Tukey FFT. Merton jump-diffusion verified by triple agreement across semi-analytic, FFT, and Monte Carlo methods. Variance Gamma in progress.
- **FX Options Portfolio Risk Aggregator:** Python tool pricing a multi-currency FX options portfolio via Garman-Kohlhagen, computing PV, Delta, and Vega with Pydantic validation and 10 unit tests covering put-call parity, edge cases, and notional conversion.

HONORS AND AWARDS

- Ranked 1st in Department for Final Year Project, University of York, 2025
- Certified Kubernetes Application Developer (CKAD), Linux Foundation, 2022
- Best Final Year Project, Computing Showcase, University of Plymouth, 2020